

Task 3: Malware Detection Tool

Objective

The purpose of this project is to develop a basic malware detection tool that scans files based on hash signatures. This tool helps in identifying malicious files by comparing their hash values with known malware signatures.

Tools & Technologies Used

Python: Programming language for implementing the scanner.

Hashlib: Library for computing file hashes.

Implementation Details

: Compute File Hash

The tool reads the target file and computes its SHA-256 hash.

This hash is then compared against a database of known malicious hashes.

: Hash Signature Database

A predefined list of malware hashes is maintained within the script.

The tool checks if the computed hash matches any known malicious hash.

Process:-

1. User provides the file path as an argument.
2. The tool calculates the file's SHA-256 hash.
3. The computed hash is compared against the database.
4. If a match is found, the file is flagged as malicious.
5. Otherwise, it is considered safe.

Testing

Test Case 1: Scanning a Clean File

Input: safe_document.txt

Expected Output: "The file is safe."

Test Case 2: Scanning a Malicious File

Input: malware.exe

Expected Output: "Warning! The file is malicious."

Findings & Recommendations

Findings:

The tool successfully detects files with known malicious hashes.

Performance is fast for small file scans.

Improvements:

Expand the malware hash database.

Integrate with VirusTotal API for broader detection.

Implement real-time monitoring for directories.

Conclusion

This project demonstrates a basic malware detection tool that identifies threats using hash comparison. Further enhancements will improve detection accuracy and efficiency.